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Activity-Based Costing and Capacity

Measuring and managing the cost of capacity is at the heart of activity-based costing. In this note, we show how to incorporate both budgeted and capacity information in ABC models.

Historical (Traditional) ABC

Activity-based costing involves detailed analysis of organizational activities. For example, consider an activity, *handle customer orders*. An ABC analyst reviews historical data for the most recent period (e.g., quarter, half-year, or year) and conducts interviews and surveys to estimate the quantity of organizational expenses that can be traced to the resources (people, systems, facilities) performing the activity. Suppose the traced expenses equal \$250,000. In the next step, the analyst selects an activity cost driver for this activity, *number of customer orders*, and determines the quantity of this activity cost driver in the most recent period. Say this number is 3,500 orders. The analyst then estimates the historical cost-driver rate as \$71.43 per order (\$250,000 in expenses divided by the 3,500 orders). This rate is used to calculate the cost and profitability of individual orders and customers during the estimation period.

This historical method is objective and relatively simple to implement. To estimate activity cost-driver rates with the historical method, analysts:

1. Trace the resource expenses to activities
2. Obtain the quantitative data on the output of each activity
3. Measure how much of each activity's output was used by each product or customer

Such a calculation, while more accurate than a traditional cost calculation, is not that useful or even correct. First, the actual cost-driver rate is not calculated until after the period is over. You can readily imagine the following statement from a manager to an ABC cost analyst: "You are still a traditional accountant, looking backwards and telling me where I have been. I need a system that helps me navigate to the future, that informs me about what I should be doing, not just reporting more accurately about past operations."

For ABC to be used as an ongoing system, managers cannot wait until the end of the period to obtain cost-driver rates for calculating product and customer costs and profitability.

Professor Robert S. Kaplan prepared this note as the basis for class discussion.

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Second, the cost-driver rate is inaccurate if the resources performing the activity are operating below capacity. If the organization has the capacity to handle 5,000 customer orders during the period, not just the 3,500 that actually present themselves, then the correct cost-driver rate is considerably less than \$71.43. The rate calculated at actual utilization includes both the cost of resources used to handle each customer order as well as the cost of resources that were supplied **but not used** during the period. Both these limitations need to be overcome if ABC systems are to be used proactively and correctly.

Viewing the Future: ABC with Budgeted Expenses

Once the basic structure of an ABC model has been determined, one can use, as inputs to an ABC model, the budgeted expenses for resources in the upcoming period. In this way, activity cost-driver rates are a function of anticipated expenses. This enables cost-driver rates to be calculated, in advance, at the beginning of a period so that managers can use this information, in real time, when making decisions about products and customers.

Continuing our numerical example, assume for an upcoming period that the budgeted expenses are \$280,000 for the resources that *handle customer orders*. And suppose the expected number of customer orders during the period is 4,000 orders. The ABC model calculates a cost-driver rate of \$70 per order based on budgeted expenses and activity levels. This amount (\$70) can be charged to every customer order received during the period. Managers can use this information to make decisions about pricing, minimum order size, and order acceptance during the period. They do not have to wait until the end of the period to learn how much it costs to handle an individual order.

Working with budgeted rather than historical expenses overcomes one problem with traditional ABC. Managers now have a cost model to forecast the future, not just explain the past. They must still cope with the second limitation. Even working with budgeted data, the forecasted activity volume may be well below the quantity that could be handled by the resources supplied to perform this activity. With our example, the \$70 activity cost-driver rate may include expenses of both used and unused resources.

Measuring the Cost of Capacity Resources

If managers use either actual or forecasted activity levels to calculate cost-driver rates, they risk launching a death spiral in their organization. The resources supplied to perform most indirect and support activities are fixed in the short run. If activity levels decline—perhaps because of a general slowdown in economic activity or loss of a major customer—then the cost-driver rate will increase (a simple arithmetic calculation since expenses, the numerator of the calculation, remain the same, while the cost-driver quantity, the denominator, declines). If the now higher cost-driver rate is used for decisions about pricing, discounting, and order acceptance, the company may set a higher reference price to offset the higher cost-driver rate. But such actions could lead to even lower activity levels as customers balk at the attempt to recover excess capacity costs at their expense. If these lower activity levels are then fed back into the cost-driver rate calculation, an even higher cost-driver rate gets calculated the next period, reinforcing the vicious cycle of losing more business and recalculating higher cost-driver rates.

The use of forecasted activity levels not only risks such a death spiral, the calculation is conceptually incorrect. If, as we have assumed, the resources supplied to perform an activity, such as *handle customer orders*, are essentially fixed in the short run, we need to obtain an additional and very important new piece of information: How many customer orders can be handled during the period

by the supplied resources? This new information represents the practical capacity of the resources for this activity, the largest number of customer orders that can be handled without creating unusual delays, forcing overtime work, or requiring additional resources to be supplied. Suppose, for purposes of illustration, that the practical capacity for this activity is 5,000 customer orders per period.¹ In this case, the correct cost-driver rate is \$56 per order, not the \$70 per order calculated using the forecasted activity volume. Why is \$56 “more correct” than the \$70?

Managers, through past or recent budgeting decisions, have authorized a supply of resources expected to cost \$280,000 for performing the *handle customer order* activity. What have they received from this authorization? They have obtained a **capacity** to handle 5,000 customer orders. Assuming that each customer order requires approximately the same quantity of resources to handle, then about \$56 worth of resources is used each time a customer order is handled. This number represents the basic efficiency of the order-handling processes, \$56 worth of resources for each order. If, in a period, only 4,000 orders are received, the efficiency of the activity remains about the same, requiring only \$56 worth of resources to process each order. The people and other resources required to perform this process do not suddenly get less efficient (\$70 per order) just because fewer orders are received in a period. The lower number of orders received translates into using less than the full quantity of resources supplied during the period. Because of contracts and commitments (explicit and implicit) made to the resources performing this activity, the supply of resources cannot be lowered in the short run in response to the expected lower activity level (that is what we mean by a “fixed” cost). Even if they can, managers may choose to retain the current level of resource supply in order to handle higher expected order volumes in the future.

In either case, the cost-driver rate should reflect the underlying efficiency of the process—the cost of handling a customer order—and this is measured better by using the capacity of the resources supplied as the denominator when calculating the rate. The numerator in the cost-driver rate calculation represents the costs of supplying resource capacity to do work. The denominator should match the numerator by representing the quantity of work the resources can perform.

Committed and Flexible Resources

Organizations contract to supply most of their resources before they are actually used. The organization makes a commitment or actual cash outlay to acquire such resources that will be used for current and future activities. We refer to these resources as *committed resources*. Consider the acquisition of buildings and equipment. Such a transaction leads to an expense being recognized in each period during the useful life of these resources. The expense of supplying the resource is incurred, each period, independent of how much of the resource is used. As another example, the organization often enters into explicit contracts to obtain the services from resources for several periods in the future. It leases building and equipment or enters into take-or-pay contracts to guarantee access to energy or strategic materials. The cash payments and associated expenses will be independent of the actual quantity of usage of such resources in any period.

Perhaps the most important contracts are the implicit contracts (and occasionally explicit contracts, especially with top executives or a unionized workforce) to maintain employment levels despite short-term downturns in activity levels. Engineers, purchasing managers, production

¹ It is not important that such a practical capacity be estimated exactly. Whether the capacity is actually 4,800 or 5,200 is less important than choosing a particular number to approximate the practical capacity and to continue to use this number each time the cost-driver rate is calculated until the organization has adequate evidence to obtain a better estimate. Distinguishing between practical capacity and normal volume only becomes an issue when normal volume is substantially below practical capacity. As in many cost-measurement issues, it is more important to be approximately correct than precisely wrong.

supervisors, sales and marketing managers, and the salaried workforce remain on the payroll even when the short-term demand for their services declines. The spending and accrued expenses for such employees remain constant, independent of the quantity of work performed by the employees.

In each of these cases, the organization acquires service capacity before the actual demands for the service are realized. Consequently, the expenses of supplying the service capacity are incurred whether the resources are used or not. The independence in the short-run supply and expenses of these resources has led many to label these as *fixed costs*.

From this perspective, variable costs represent only those resources that the organization acquires as needed. Such resources, which we refer to as *flexible resources*, are typically acquired from outside suppliers; examples include materials, energy, telecommunications services, temporary workers hired on a daily basis, overtime pay, and employees paid on a piecework basis. For these resources, the organization acquires only what it needs to meet short-term demands, so the cost of acquiring these resources equals the cost of using them. These resources have no unused capacity; whatever is supplied is used, or conversely, whatever is needed is supplied.

Activity-based systems incorporate the costs of all resources, both flexible and committed, used by activities and products. ABC systems recognize that almost all organizational costs (other than those of flexible resources) are not variable in the short run just because demand fluctuates up and down. Rather, committed costs become variable costs over longer time periods via a two-step procedure. First, demands for the resources supplied changes because of changes in activity levels. In the second step, the organization changes the supply of committed resources, either up or down, to meet the new level in demand for the activities performed by these resources.

When the capacity of existing resources is exceeded, the pain is obvious through shortages, increased pace of activity, delays, or poor-quality work. Such shortages can occur on machines, the usual case that people think about. But the ABC approach makes clear that such shortages can occur for the people resources performing support activities, such as designing, scheduling, ordering, maintaining, and handling products and customers. Companies, facing such shortages, move to the second step of making committed costs variable: they relieve the pain by spending more to increase the supply of resources and relieve the bottleneck.

Demands for resources can also decline. Should the demands for committed resources decrease from managers' actions, little immediate spending improvement occurs. The reduced demand for organizational resources **lowers the cost of resources used** (by products, services, and customers), but the decrease is offset by an equivalent **increase in the cost of unused capacity**.

For committed costs to become variable in the downward direction, the organization must manage the unused capacity of these resources out of the system. At that point, and only at that point, will the costs of resources supplied start to decrease. What makes a resource cost "variable" in a downward direction is not inherent in the nature of the resource; it is a function of management decisions—first to reduce the demands for the resource and thereby create unused capacity, and, second, to lower the spending on the resource no longer needed.²

Even actions taken to improve the efficiency of activities require an understanding of how unused capacity gets created and subsequently managed. Suppose that a continuous (TQM) or discontinuous

² Managers do not seem to have any problem recognizing how costs are "variable" in an upward direction. Examination of past history will usually reveal how organizational spending has increased to cope with increased variety and complexity of operations. It's the mechanism for costs to head in the downward direction that seems to have eluded most managers and academics.

(reengineering) improvement initiative is launched to reduce waste and non-value-added tasks in the customer order handling activity. Perhaps a modest technology investment, such as electronic data interchange (EDI), is made to help employees be more productive. Such initiatives typically result in expanding the capacity to perform work (since the amount of time and resources devoted to non-productive work will be sharply reduced or eliminated). The practical capacity rises, say to 7,000 customer orders per month. But expenses remain the same; they may even rise slightly if any technology investments are made. People have been hired; they continue to come to work; they use their desks, computers, and telephones; they continue to take coffee breaks, vacations, and sick leave; and they accumulate time to pensions.

Has the cost of handling customer orders been reduced by the TQM or reengineering initiative? Absolutely! Before, the cost of handling a customer order required \$56 worth of resources. After the initiative, a customer order can be handled with only \$40 worth of resources ($\$280,000/7,000$), perhaps a little higher to cover any new technology investment. But suppose that only 4,000 orders are expected the next period. How much should managers expect to **spend** next period on resources devoted to handling customer orders? The answer, of course, is \$280,000. If business activity goes as planned, the expenses assigned to the actual orders received will decrease to \$160,000 ($4,000 @ \40) from the previous level of \$196,000. This will signal that the order-handling cost has been reduced because of the TQM/reengineering initiative. But the cost of unused capacity rises to \$120,000 ($3,000 @ \40) since all the resources previously supplied are still being paid for.

Efficiency initiatives lead to reduced spending only if managers eliminate or redeploy the unused resource capacity they have created. The example clearly shows that applying TQM or reengineering to resources that already exceed the supply needed to perform the forecasted level of activities will likely produce only more excess capacity. The ABC system signals those resources that are currently at or expected to soon reach capacity constraints. Initiatives can then be focused on improving the efficiency of the activities performed by these constraining resources. The ABC system also signals where unused capacity already exists or will be created after improvement activities or after decisions about reducing resource demands by products, services, and customers. Such a signal directs managers' attention to eliminating the unused capacity by taking one or both of the following actions:

1. Reduce the supply of resources for performing the activity.
2. Increase the quantity of revenue-generating activities performed by the resources.

Often, however, organizations create unused capacity but do nothing about it. They keep existing resources in place, even though the demands for the activities performed by the resources have diminished substantially. And they fail to find new revenue opportunities that could be handled, without incremental expenses, by the resources already in place. In this case, and only in this case, will the organization not receive any benefit from its decisions. But the failure to capture benefits from operational or strategic decisions is not due to costs being intrinsically "fixed." Rather, the failure occurs because managers are unwilling or unable to exploit the unused capacity they have created. The costs of these resources is "fixed" only if managers do not exploit the opportunities from the unused capacity they helped to create.

Measuring Capacity

We have illustrated the calculation of capacity-based cost-driver rates with a simple numerical example. In practice many complications arise when attempting to measure practical capacity. Here, we discuss just a few of these and provide rules of thumb for coping with them.

What Is Practical Capacity?

Practical capacity can be studied analytically in some depth or estimated somewhat arbitrarily. The accurate analytic approach starts with measuring theoretical capacity and subtracting the time not available for actual work; for machines, the time not available is reserved for preventive maintenance and minor repairs. The arbitrary approach assumes that practical capacity is a specified percentage, say 80% or 85%, of theoretical capacity. That is, if an employee or machine normally can work 40 hours per week, practical capacity could be assumed to be 32 hours per week, allowing 20% of personnel time for training, breaks, communication, and reading; for machines, the allowance covers downtime due to maintenance, repair, and scheduling fluctuations.

For a simple but less arbitrary approach, the analyst can review the time series of past activity levels. For example, look at the number of customer orders handled over the past 12 or 24 months and identify the month with the maximum number of orders. Check whether for that period the work was handled without excessive delays, poor quality, overtime, or stressed employees. If not, as a starting point, one can use that maximum number as the estimate of the capacity of the resources performing that activity. The analysis is not greatly sensitive to small errors in estimating parameters. It is better to make a reasonable estimate of capacity and continue to use that number as the denominator volume in calculating the cost-driver rate than to introduce arbitrary fluctuations by using last period's actual or next period's expected activity level when calculating cost-driver rates.

Lumpiness in Acquiring Capacity

Suppose unused capacity arises from the lumpiness of asset acquisition. The company wants a machine to process 800,000 units per year, but the smallest machine with this minimum capacity could produce 1 million units per year. The company acquires a machine with a practical capacity of 1 million units per year, but because of demand constraints or constraining resources elsewhere in the production process, it anticipates producing only a maximum of 800,000 units per year. If the decision to acquire the machine anticipated this maximum usage, then the practical capacity should be measured as 800,000, not 1 million. Should external demand become higher than anticipated, or if internal bottlenecks elsewhere in the plant get relieved so that output climbs unexpectedly to 900,000 units per year, then, of course, the denominator volume can get reset to the higher figure.

A variation of this theme occurs when a machine is acquired whose annual cost (rental or amortization of purchase cost plus operating cost) is, say, \$120,000 per year. Demand is expected to build over time, starting with a low value of 100,000 units the first year, 200,000 units in year two, and reaching 400,000 units in year three and thereafter. The naïve approach would estimate the first-year cost at \$1.20 per unit, the second-year cost at \$0.60 per unit, and the third-year cost at \$0.30 per unit. These calculations suggest that the machine is four times more efficient in year three than in year one, when, in fact, its efficiency is identical for the three years. Only the volume of business is changing, not the efficiency. Two approaches seem plausible in this situation. The first uses the practical capacity (400,000 units per year) as the denominator volume so that machine costs would be \$0.30 independent of the year in which a unit was produced. The unused capacity of \$90,000 in year one is anticipated in the capital-budgeting decision and could be charged either to the investment account or written off as unused capacity. It should not be incorporated into the cost of the resources used (25% of capacity) by year-one production.

Alternatively, the operating costs over the lifetime of the equipment could be assigned to anticipated lifetime production (sum of production units in years one, two, three, up through the maximum economic life of the equipment). This also produces a constant cost per unit produced,

independent of which year production of the unit occurs, but does not lead to an unused capacity amount in any year (unless actual production is less than anticipated).

Seasonal and Peak-Load Capacity

The next extension is how to handle seasonal or peak-load demands. Suppose (see **Exhibit 1**) the company has demand for 8,000 units of output in months 1–4 of the year and demand for only 4,000 units of output for months 5–12. One can think of selling ice cream or soda pop, with peak demand in the four warmest months and low demand in the cool months; or supplying audit services, where peak demand occurs from January to April because most companies have fiscal year-ends on December 31; or phone companies, which supply peak demand during the eight hours of the normal business day and lower (consumer) demand in the other 16 hours. Suppose the company enjoys economies of scale in acquiring capacity so that it costs the company \$4,000 per month to meet the low demand of 4,000 units of output per month but \$6,000 if it meets the peak demand of 8,000 units per month. The capacity must be acquired for the entire year; the company cannot gear up for the peak period and then scale back the resources during the slack-demand period.

The company decides to meet the peak demand and commits to resources costing \$6,000 per month for the entire year. How should the \$72,000 cost be assigned to the annual output of 64,000 units ($4 \times 8,000 + 8 \times 4,000$)? Two simplistic (and incorrect) approaches assign the cost uniformly to outputs.

1. Annual calculation: $\$72,000 / 64,000 \text{ units} = \1.125 per unit
2. Monthly calculation:
 - a. Peak period: $\$6,000 / 8,000 \text{ units} = \0.75 per unit
 - b. Slack period: $\$6,000 / 4,000 \text{ units} = \1.50 per unit

The monthly unit cost is lower in the peak period because production in this period is higher than in the slack period. This makes no economic sense; the peak-period demands required additional and costly resources to be acquired. The annual unit cost calculation does not distort by signaling lower costs in peak periods and higher costs in slack periods, but it too does not signal the higher costs (more capacity resources) required for peak-period production.

The preferred calculation recognizes that production in the peak period requires not only the capacity resources used during this period but also the cost of the resources acquired but unused during the slack period. The cost of capacity in the slack period should just be the capacity that would be required if only the slack-period demand were to be met:

$$\begin{aligned} \text{Cost in months 5–12} &= \$4,000 \text{ per month} / 4,000 \text{ units per month} &&= \$1 \text{ per unit} \\ \text{Cost in months 1–4} &= \frac{\$6,000 * 4 + (\$6,000 - 4,000) * 8}{8,000 * 4} &&= \$1.25 \text{ per unit} \end{aligned}$$

The total cost of supplying peak capacity in the four peak months is the monthly cost of peak capacity (\$6,000 per month) plus the cost of unused capacity ($\$6,000 - 4,000$) = \$2,000 in the eight slack-demand months (or \$16,000 in total). The total-capacity cost of \$40,000 for these four months can then be divided by the total production of these four months (32,000 units) to obtain the \$1.25 per unit cost. These calculations conform better to economic reality. Production in peak periods is more expensive (\$1.25 vs. 1.00) because of the cost of the additional capacity required to handle the peak-

period demands. Production in the peak-demand period must pay for not only the costs of capacity resources it uses but also the cost of capacity resources supplied, but not used, during the slack-demand period.

Capacity to Provide Enhanced Service Quality

Suppose a telecommunications company has two types of customers. The first type needs only low-speed, moderate-quality voice transmission. The second type requires high-speed, very high-quality communication capable not only of excellent voice communications but also high-speed data and video transmission. Rather than supply two different and parallel communications channels, the company finds it cheaper to build the higher-quality but more expensive channel required to attract the second type of customer. Both customer types use the same network, but if the same average unit cost is assigned to both, the first type is being charged with costs for a higher-quality service than it is willing to pay. The decision to invest in the more expensive channel is justified by the preferences of the second type of customer and the decision of the telecommunications company to meet the demands of that customer type.

In this case, one could estimate the costs to be assigned to the first customer type by a simple mental or modeling experiment. How much would it cost to construct a basic network (low-speed, low-bandwidth communication) assuming that all customers were of type one? The unit cost of providing such a basic network is a reasonable cost assignment to type one customers. The extra cost to build the network with the high-speed, high-quality features and service can all be attributed to type two customers. So even though both customers use the same network, the less demanding customers are assigned only the costs associated with delivering the desired value proposition to them; the demanding type two customers bear all the incremental costs required to provide them with a high-quality communications network. The ability to assign costs differentially to different types of customers depends critically on being able to distinguish among the different service levels required by diverse customer segments. But such knowledge is frequently available, so such distinctions can be made in practice and the relevant costs estimated.

Assignment of Unused Capacity Costs

When activity cost-driver rates are based on practical capacity, the cost of unused capacity is not assigned down to individual products or customers. But the cost of unused capacity should not be ignored; it remains someone's or some department's responsibility. If the capacity was acquired to meet anticipated demands from a particular customer or a particular market segment, then the costs of unused capacity due to lower-than-expected demands can be assigned to the person or organizational unit responsible for that customer or segment. If the unused capacity relates to a product line, say when certain production resources are dedicated to individual product lines, then the cost of unused capacity is assigned to the individual in charge of the product line where demand failed to materialize. It should not be treated as a general cost, to be shared across all product lines. Nor should the product-line unused capacity be allocated down to individual products. That could cause some products to appear unprofitable, risking the launch of a death spiral by repricing or dropping products.

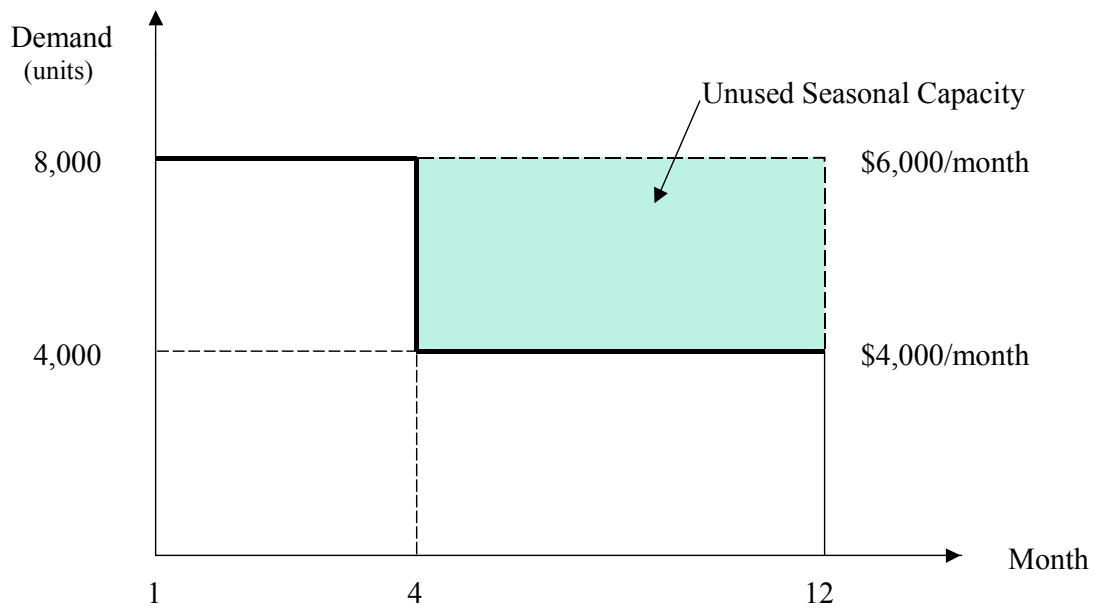
As another situation, suppose division management knew in advance that resource supply would exceed resource demand but wanted to retain existing resources for future growth and expansion. Then the cost of unused capacity would be division sustaining, assigned to the division making the decision to retain unused capacity. In making such assignment of unused capacity costs, management aims to trace the costs to the level in the organization where decisions are made that affect the

demand and supply of capacity resources. The lump-sum assignment of unused capacity costs provides feedback to managers on their decisions about demand and supply.

Summary

An organization's initial ABC system typically calculates historical cost-driver rates, based on last period's expenses for resources actually supplied and actual activity levels. But ABC should not be thought of as a historical accounting or general ledger system. It can and should be used pro-actively to estimate the costs of activities that will be performed in current and future periods. This enables decisions to be made that can influence future cost incurrence, not just to better assign past costs. For this purpose, activity cost-driver rates should be calculated using budgeted expense data for the forthcoming period.

In addition to using budgeted expense information, the rates should be calculated using the practical capacity of the resources supplied. In addition to giving a more accurate estimate of the cost-driver rate, the use of practical capacity enables ABC to distinguish between the cost of resources used during the period versus the cost of unused resources. This distinction provides a powerful signal for management as it contemplates decisions about process improvement, products, customers, and investments in new capacity.

Exhibit 1 Seasonal Capacity

Source: Casewriter.